Raghav Mangal

ce23btech11044@iith.ac.in | raghavmangal76@gmail.com | +91 9256381350 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.81
Jayshree Periwal High School	Class XII (CBSE)	2023	93.00%
Ryan International School	Class X (CBSE)	2021	94.00%

SKILLS

Programming Languages: C++, C, Python, SQL

AI/ML : LoRA/PEFT Fine-tuning, Vision Transformers, Model Quantization, Unsloth, Prompt Engineering

Software Engineering: Data Structures & Algorithms, OOPs, DBMS, Operating System, REST APIs, Git

Cloud & Data Engineering: AWS (S3, SageMaker), Snowflake, Data Pipeline Design

Tools & Platforms: JupyterHub, VS Code, Kaggle, Git, Excel

WORK EXPERIENCE

Applied AI Intern -- Piramal Finance

Computer Vision and Document Intelligence - End to End ownership

- Developed a **VLM-based document image classification and verification pipeline** for document redaction, Aadhaar masking, classifying documents as masked, unmasked, or absent to automatically validate data masking.
- **Data Pipeline** - Built a multithreaded ingestion pipeline to stream **1.1M+ documents** from Kanaja API into AWS S3 with supporting data fetched via Snowflake, followed by data preprocessing, resulting in a final curated dataset.
- **Model Development:** Built an initial prototype using **GPT-4o API,** followed by fine-tuning VLMs (**Gemma-4 12B-IT and Qwen-3VL 8B**) using LoRA on SageMaker, quantizing to FP8 for efficient inference, and further exploring Vision encoder architectures (ViT/DINOv2) on JupyterHub for faster classification.
- **Optimization:** Improved model performance through systematic experiments on prompt tuning, image properties, tokenization and LoRA configurations -- raising accuracy from **63% baseline accuracy** to **99.3% accuracy.**
- **Deployment:** Designed the inference pipeline routing all documents through ViT first (**97.45% accuracy, 3.4ms/doc**), escalating only low-confidence or "masked" predictions to VLM (**98.8% accuracy, 400ms/doc**)

PROJECTS

Predictive Distributed-System Reliability Platform

- Built a real-time distributed reliability platform that predicts infrastructure failures hours before they happen, using time-series trend analysis.
- Designed a time-series ingestion and prediction pipeline - simulated exporters -> streaming ingestion -> SQLite/TimescaleDB-ready storage -> FastAPI , that converts metric trends into actionable alerts.
- Implemented trend-based forecasting that estimates time-to-failure with confidence scores, cross-checking leading indicators against confirming signals to reduce false alarms.

AI Code Review System

- Built an AI-powered code review system that analyzes GitHub pull request diffs and generates categorized feedback using LLM-based static+semantic analysis, reducing manual review time.
- Designed a diff-parsing and context-retrieval pipeline that extracts surrounding code context and repo conventions to improve LLM review accuracy, integrated with GitHub for automated PR comments.
- Combined static analysis tools with LLM-based review to cross-validate findings and reduce false positives in automated code review.
- Implemented a scoring mechanism to rank flagged issues, improving signal-to-noise ratio for reviewers and enabling prioritized triage of PR feedback.

Content & Event Management Platform

- Developed a full-stack web platform using Spring Boot REST APIs, React, and PostgreSQL, implementing authentication, CRUD operations, search, and role-based access control.
- Designed modular backend services following OOP and RESTful architecture, with validation, exception handling, and reusable service components.
- Improved API performance using Redis caching and database indexing, reducing repeated database queries and improving response latency.
- Containerized the application using Docker and deployed backend services on AWS.

SCHOLASTIC ACHIEVEMENTS

- Solved **700+ Problems** on LeetCode and Geeks for Geeks.
- Solved **250+ problems** on Codeforces and CodeChef.
- Achieved a **rating of 1700+** on CodeChef.
- Achieved a **rating of 1500+** on CodeForces.

POSITIONS OF RESPONSIBILITY

- **E-cell** (Entrepreneurship cell), **Corporate Relations and Finance Manager**, IIT Hyderabad: Lead a 4-member team handling corporate outreach, sponsorship negotiations and budgeting across **100+ companies.**
- **Elan N Vision** (Technical and Cultural fest), **Techy Assistant Coordinator**, IIT Hyderabad: Coordinated across teams to plan and execute large-scale technical events.